Saurabh Bhandari

Contact Information

Email: sbhandari52@uchicago.edu
Website: https://sbstats.github.io/
Phone: 352-219-2040

Education

08/2020–08/2025 **Ph.D., Statistics** - University of Florida, Gainesville, Florida
Advisor: Dr. Michael J. Daniels

08/2016–05/2020 **B.S., Computer Science** - Troy University, Troy, Alabama
Second Major: Mathematics

Employment

09/2025–Present **Postdoctoral Scholar** - University of Chicago, Chicago, IL
Supervisors: Dr. Yuan Ji and Dr. Brian C.-H. Chiu

05/2023–08/2023 **Ph.D. Intern (Biostatistics)** - Regeneron Pharmaceuticals, Tarrytown, NY
Supervisor: Dr. Chenguang Wang

Reseach Interests

- **Methodology:** Causal inference with observational data, machine learning, Bayesian nonparametric modeling, joint analysis of longitudinal and survival data.
- **Applications:** Biomedical data science, oncology, public health, digital communications.

Publications

Published/Under peer-review

1. **Bhandari, S.**, Daniels, M.J., Josefsson, M., Lloyd-Jones D.M., Siddique J. A Bayesian semi-parametric approach to causal mediation for longitudinal mediators and time-to-event outcomes with application to a cardiovascular disease cohort study. *Biostatistics* 26.1 (2025): kxaf027. (https://doi.org/10.1093/biostatistics/kxaf027)
2. **Bhandari, S.**, Derman, B., Zhang Z., Kowitwanich K., Cui X., Bhatti, P., Jakubowiak, A., He, C., Zhang, W., Ji, Y., Chiu, B. Pooled analysis of circulating cell-free DNA 5-hydroxymethylcytosine signatures associated with multiple myeloma using Bayesian machine learning. *Under revision.*
3. **Bhandari, S.**, Li F., Daniels, M.J. A Bayesian machine learning approach to direct and spillover effects in cluster-randomized trials with noncompliance. *Under review.* (arXiv: https://arxiv.org/abs/2610.27425)

4. **Bhandari, S.**, Kar, W., Daniels, M.J., Karmakar, B. Causal mediation analysis for longitudinal data in the presence of treatment non-compliance and multiple mediators. *Under review.* (arXiv: https://arxiv.org/abs/2608.15775)
5. **Bhandari, S.**, Bhatti, P., Chiu, B., Ji, Y. A semiparametric Bayesian machine learning model to analyze high-dimensional epigenetic signatures associated with multiple myeloma. *Under review.* (arXiv: https://arxiv.org/abs/2605.20143)

Manuscripts in preparation

1. **Bhandari, S.**, Chiu, B., Ji, Y. TS-BART: Tree-selection Bayesian Additive Regression Trees.
   - An R package (on CRAN) is being developed for this project.
2. **Bhandari, S.**, Daniels, M.J., Siddique J. Causal mediation analysis for longitudinal and survival data in continuous time using Bayesian non-parametric joint models. (arXiv: https://arxiv.org/abs/2506.20058)
3. **Bhandari, S.**, Daniels, M.J., Wang, C. Integrating tumor burden with survival outcome for treatment effect evaluation in oncology trials. (arXiv: https://arxiv.org/abs/2506.07387)
4. **Bhandari, S.**, ..., Ji, Y., Chiu, B. Prognostic significance of a cell-free DNA 5-hydroxymethylcytosine in newly diagnosed multiple myeloma.

Talks and Presentations

Invited talks

4. **The 2026 ISBA World Meeting**. Nagoya, Japan. Causal mediation analysis for longitudinal and survival data in continuous time using Bayesian non-parametric joint models. Jun-Jul 2026.

3. **University of Illinois Chicago (UIC) Biostatistics Seminar Series**. Virtual. A Bayesian semi-parametric approach to causal mediation for longitudinal mediators and time-to-event outcomes. Feb 2026.

2. **ICSA Applied Statistics Symposium 2025**. Storrs, CT. Integrating tumor burden with survival outcome for treatment effect evaluation in oncology trials. Jun 2025.

1. **CMStatistics 2023**. Berlin, Germany. A Bayesian semi-parametric approach to causal mediation for longitudinal mediators and time-to-event outcomes. Dec 2023.

Topic-contributed talk

2. **JSM 2026**. Boston, MA. Causal mediation analysis for longitudinal and survival data in continuous time using Bayesian non-parametric joint models. Aug 2026.

1. **JSM 2025**. Nashville, TN. Causal mediation analysis for longitudinal and survival data in continuous time using Bayesian non-parametric joint models. Aug 2025.

Contributed talks

3. **IBC 2024**. Atlanta, GA. A Bayesian semi-parametric approach to causal mediation for longitudinal mediators and time-to-event outcomes. Dec 2024.

2. **Mark C. K. Yang Mentor Ceremony Event 2024**. Department of Biostatistics, University of Florida, Gainesville, FL. A Bayesian semi-parametric approach to causal mediation for longitudinal mediators and time-to-event outcomes. Nov 2024.

1. **JSM 2024**. Portland, OR. A Bayesian semi-parametric approach to causal mediation for longitudinal mediators and time-to-event outcomes. Aug 2024.

Poster presentations

2. **Society of Epidemiologic Research (SER) Annual Meeting 2026**. Phoenix, AZ. Pooled analysis of cell-free DNA derived 5-hydroxymethylcytosine signatures and risk of multiple myeloma using Bayesian machine learning. Jun 2026.

1. **Best of Statistical Science (BOSS) 2025 Workshop**. Department of Statistics, Texas A&M University, College Station, TX. Causal mediation analysis for longitudinal and survival data in continuous time using Bayesian non-parametric joint models. Apr 2025.

Teaching Experience

Winter 2026	**University of Chicago** Class: PHS 31001- Epidemiological Methods	Position: Guest Speaker
Fall 2024	**University of Florida** Class: STA 3024- Introduction to Statistics II	Position: Instructor
Summer 2022	**University of Florida** Class: STA 3024- Introduction to Statistics I	Position: Instructor
08/2020–06/2022	**University of Florida**	Position: Graduate TA

Service

Journal Referee	*Biometrics* (3), *Statistics in Medicine* (1)
Chair and organizer	Invited paper session *"Modern causal inference and machine learning methods for cluster-randomized trials"* at JSM 2027, Chicago, IL.
Judge	ASA Datafest 2023, University of Florida Department of Biostatistics, Gainesville, FL.

Honors and Awards

Jun 2026	International Society for Bayesian Analysis (ISBA) Program Council *Junior Travel Award for Attendance at the 2026 ISBA World Meeting in Nagoya, Japan*
Jun 2025	College of Liberal Arts and Sciences at the University of Florida *Graduate Travel Award*
Nov 2024	Department of Biostatistics at the University of Florida *Mark C. K. Yang Mentor Ceremony Event 2024 Student Presentation Award*
Dec 2023	College of Liberal Arts and Sciences at the University of Florida *Graduate Travel Award*
May 2020	Troy University Mathematics Department *Pi Mu Epsilon Excellence In Leadership Award - 2020*

References

1. **Dr. Michael J. Daniels**, Professor and Chair, Department of Statistics, University of Florida
   Email: daniels@ufl.edu
2. **Dr. Brian C.-H. Chiu**, Professor, Department of Public Health Sciences, University of Chicago
   Email: bchiu@bsd.uchicago.edu
3. **Dr. Yuan Ji**, Professor, Department of Public Health Sciences, University of Chicago
   Email: yji@bsd.uchicago.edu
4. **Dr. Bikram Karmakar**, Assistant Professor, Department of Statistics, University of Wisconsin-Madison
   Email: bkarmakar@wisc.edu

November 7, 2026